The National Institute of Engineering

Scheme of Teaching & Examination - 2022

Effective from the Academic year 2023-24

Department: Computer Science and Engineering (AI and ML)

B.E. 2022 Admitted Batch

Semester: IV

Sl. No.	Type of Course	Course Code	Course Title	Teaching Department (TD)	Question Paper setting Board (PSB)	Teaching Hrs/Week				Examination				Credits
						L	T	P	S	Duration in Hours	CIE Marks	SEE Marks	Total Marks	
1	PCC/BSC	BCS401	Analysis & Design of Algorithms	TD: CS	PSB: CS	3	0	0	0	3	50	50	100	**3**
2	IPCC	BAI402	Artificial Intelligence	TD: CS	PSB: CS	3	0	2	0	3	50	50	100	**4**
3	IPCC	BCS403	Database Management Systems	TD: CS	PSB: CS	3	0	2	0	3	50	50	100	**4**
4	PCCL	BCSL404	Analysis & Design of Algorithms Lab	TD: CS	PSB: CS	0	0	2	0	2	50	50	100	**1**
5	ESC	BCS405x	ESC/ETC/PLC	TD: CS	PSB: CS	2	2	0	0	3	50	50	100	**3**
6	AEC/SEC	BCS456x	Ability Enhancement Course (AEC) - IV	TD: CS	PSB: CS	If the course is a Theory 1	0	0	0	1	50	50	100	**1**
						If the course is a Laboratory 0	0	2	0	2				
7	BSC	BBOK407	Biology for Engineers	TD: Chem	PSB: Chem	2	0	0	0	2	50	50	100	**2**
8	UHV	BUHK408	Universal Human Values and Professional Ethics	TD: CS	PSB: CS	1	0	0	0	-	50	-	50	**1**
9	MC	BNSK459	National Service Scheme (NSS)	NSS Coordinator		0	0	2	0	-	100	-	100	**0**
		BPEK459	Physical Education (PE) (Sports & Athletics)	PED										
		BYOK459	Yoga	Yoga Teacher										
										Total	**550**	**350**	**850**	**19**

PCC: Professional Core Course, PCCL: Professional Core Course laboratory, UHV: Universal Human Value Course, MC: Mandatory Course (Non-credit), AEC: Ability Enhancement Course, SEC: Skill Enhancement Course, L: Lecture, T: Tutorial, P: Practical: Skill Development Activity, CIE: Continuous Internal Evaluation, SEE: Semester End Evaluation. K :This letter in the course code indicates common to all the stream of engineering. ESC: Engineering Science Course, ETC: Emerging Technology Course, PLC: Programming Language Course

Engineering Science Course (ESC/ETC/PLC)			
BCS405A	Discrete Mathematical Structures	BAI405C	Optimization for Machine Learning
BAI405B	Metric Spaces	BAI405D	Algorithmic Game Theory
Ability Enhancement Course -IV			
BDS456A	Scala	BDS456C	MERN
BDS456B	MongoDB	BCS456D	Technical writing using LATEX

Code: BCS401 **Course: Analysis and Design of Algorithms**
Credits: 3 **L:T:P - 3:0:0**
SEE: 50 Marks **CIE: 50 Marks**
SEE Hours: 03 **Max. Marks: 100**

Prerequisites if any	Recurrence Relations, Data Structures
Learning objectives	• To learn the methods for analyzing algorithms and evaluating their performance. • To demonstrate the efficiency of algorithms using asymptotic notations. • To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound. • To learn the concepts of P and NP complexity classes

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Apply asymptotic notational method to analyze the performance of the algorithms in terms of time complexity.	Understand, Analyze
CO2	Demonstrate divide & conquer approaches and decrease & conquer approaches to solve computational problems.	Apply
CO3	Make use of transform &conquer, dynamic programming and greedy approaches to solve the given real world or complex computational problems.	Apply
CO4	Illustrate backtracking, branch & bound and P,NP and NP Complete problems	Apply

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	2	2	1	-	-	-	-	-	-	-	-	1		3	2
CO2	3	2	3	-	-	-	-	-	-	-	-	1		3	2
CO3	2	2	3	-	-	-	-	-	-	-	-	1		3	2
CO4	2	2	3	-	-	-	-	-	-	-	-	1		3	2

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
	Module – 1			
1	**INTRODUCTION:**			
1.1	What is an Algorithm?	1	0	0
1.2	Fundamentals of Algorithmic Problem Solving.	2	0	0
	FUNDAMENTALS OF THE ANALYSIS OF ALGORITHM EFFICIENCY:			
1.3	Analysis Framework	1	0	0
1.4	Asymptotic Notations and Basic Efficiency Classes	1	0	0

1.5	Mathematical Analysis of Non recursive Algorithms	1	0	0
1.6	Mathematical Analysis of Recursive Algorithms.	1	0	0
	BRUTE FORCE APPROACHES:			
1.7	Selection Sort	1	0	0
1.8	Sequential Search	1	0	0
	Module – 2			
	BRUTE FORCE APPROACHES (contd..):			
2.1	Brute Force String Matching	1	0	0
2.2	Exhaustive Search (Travelling Salesman problem)	1	0	0
2.3	Exhaustive Search (Knapsack Problem).	1	0	0
	DECREASE-AND-CONQUER:			
2.4	Topological Sorting.	1	0	0
	DIVIDE AND CONQUER:			
2.5	Merge Sort	1	0	0
2.6	Quick Sort	1	0	0
2.7	Strassen's Matrix Multiplication.	2	0	0
	Module – 3			
	TRANSFORM-AND-CONQUER:			
3.1	Balanced Search Trees	2	0	0
3.2	Heaps and Heapsort.	2	0	0
	SPACE-TIME TRADEOFFS:			
3.3	Sorting by Counting: Comparison counting sort	2	0	0
3.4	Input Enhancement in String Matching: Horspool's Algorithm.	2	0	0
	Module – 4:			
	DYNAMIC PROGRAMMING:			
4.1	Three basic examples	1	0	0
4.2	The Knapsack Problem and Memory Functions	2	0	0
4.3	Warshall's and Floyd's Algorithms.	2	0	0
	THE GREEDY METHOD:			
4.4	Prim's Algorithm	1	0	0
4.5	Kruskal's Algorithm	1	0	0
4.6	Dijkstra's Algorithm	1	0	0
	Module – 5			
	LIMITATIONS OF ALGORITHMIC POWER:			
5.1	Decision Trees	1	0	0
5.2	P, NP, and NP-Complete Problems.	3	0	0
	COPING WITH LIMITATIONS OF ALGORITHMIC POWER:			
5.3	Backtracking (n-Queens problem)	1	0	0
5.4	Backtracking (Subset-sum problem)	1	0	0
5.5	Branch-and-Bound (Travelling Salesman Problem),	1	0	0
	Total No. of Lecture Hours	**40**	-	-
	Total No. of Tutorial Hours		**00**	-
	Total No. of Practical Hours			**00**

Textbook

1. Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian),
2017, Pearson.

ESTD : 1946

Module-1:

Chapter 1 (Sections 1.1,1.2)
Chapter 2(Sections 2.1,2.2,2.3 (Example 2 & Example 4 Only) ,2.4 (Example 2 Only))
Chapter 3(Section 3.1(only Selection sort),3.2(only sequential search)

Module-2:

Chapter 3(Section 3.4),
Chapter 4 (Sections 4.1,4.2)
Chapter 5(Section 5.1,5.2, 5.4(only Strassen's matrix multiplication)

Module-3:

Chapter 6 (Sections 6.3,6.4)
Chapter 7 (Sections 7.1,7.2)

Module-4:

Chapter 8(Sections 8.1,8.2,8.4)
Chapter 9 (Sections 9.1,9.2,9.3)

Module-5:

Chapter11 (Section 11.2, 11.3)
Chapter 12 (Sections 12.1,12.2(only Travelling Salesman Problem)

Reference books

1. Computer Algorithms/C++, Ellis Horowitz, SatrajSahni and Rajasekaran, 2nd Edition, 2014, Universities Press.
2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
3. Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)

Online Resources:

- Design and Analysis of Algorithms: https://nptel.ac.in/courses/106/101/106101060/

ESTD : 1946

Code: BAI402 **Course: Artificial Intelligence**
Credits: 4 **L: T:P - 3:0:2**
SEE: 50 Marks **CIE: 50 Marks**
SEE Hours: 3 **Max. Marks: 100**

Prerequisites if any	None
Learning objectives	1. Develop an understanding of the historical context of AI and its fundamental principles. 2. Acquire proficiency in applying basic AI principles to solve problems effectively. 3. Familiarize oneself with the methodologies of inference, perception, knowledge representation, and learning in AI.

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Utilize fundamental AI principles to address problems involving problem-solving, inference, knowledge representation, and learning.	Understanding
CO2	Evaluate search and inference algorithms within the context of problem-solving	Apply
CO3	Exhibit understanding of reasoning, uncertainty, and knowledge representation for tackling real-world problems.	Apply
CO4	Execute experiments to address problems utilizing AI techniques.	Analyze

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	2		2	2
CO2	-	3	-	-	2	-	-	-	-	-	-	2		2	2
CO3	-	-	2	2	-	-	-	-	-	-	-	2		2	2
CO4	-	-	-	3	3	-	-	-	-	-	-	2		2	2

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
Module – 1: Introduction to AI				
1.1	**Introduction:** What is AI? Foundations of AI.	2	-	-
1.2	History of Artificial Intelligence	1	-	-
1.3	The State of the Art	1	-	-
1.4	**Intelligent Agents:** Agents and environment	1	-	1
1.5	Concept of Rationality	1	-	-
1.6	The nature of environments	1	-	-
1.7	The structure of agents	1	-	-
Module – 2: Problem solving based on searching				
2.1	Problems solving Agents, Example problems	2	-	-
2.2	Searching for solutions	1	-	-
2.3	Uniformed Search strategies – Uniform cost search	1	-	-
2.4	Breadth First Search, Depth First Search	2	-	1
2.5	Depth Limited Search	1	-	-
2.6	Iterative Deepening Depth First	1	-	1
Module – 3: Heuristic Search Strategies				
3.1	Heuristic functions	1	-	1
3.2	Greedy best first search	1	-	-
3.3	A* algorithm	1	-	1
3.4	Local Search & Optimization: Hill Climbing	1	-	-
3.5	Genetic Algorithms	2	-	-

Module – 4: Logical Agents, First Order Logic, Inference in First Order Logic				
4.1	**Logical Agents:** Knowledge–based agents	1	-	-
4.2	The Wumpus world, Logic	1	-	1
4.3	Propositional logic	1	-	-
4.4	Reasoning patterns in Propositional Logic	1	-	1
4.5	**First Order Logic:** Representation Revisited, Syntax and Semantics of First Order logic	2	-	-
4.6	Using First Order logic.	1	-	-
4.7	**Inference in First Order Logic:** Propositional Versus First Order Inference	1	-	-
4.8	Unification, Forward Chaining	1	-	1
4.9	Backward Chaining, Resolution.	1	-	1
Module – 5: Quantifying Uncertainty				
5.1	Acting under Uncertainty	1	-	-
5.2	Probability Notation	2	-	-
5.3	Inference using Full Joint Distributions	1	-	1
5.4	Independence	1	-	-
5.5	Baye's Rule and its use	2	-	-
5.6	Wumpus World Revisited	1	-	-
	Total No. of Lecture Hours	**40**	-	-
	Total No. of Tutorial Hours		**00**	**-**
	Total No. of Practical Hours			**10**

Integrated Lab Programs:

Sl. No.	COs	Programs
1.	CO1	Write a program to implement Vacuum Cleaner agent for two rooms
2.	CO2	Implement and Demonstrate Depth First Search Algorithm on Water Jug Problem
3.	CO2	Implement Iterative deepening search algorithm
4.	CO2	Solve 8 puzzle problem using BFS algorithm
5.	CO2	Implement A* Search algorithm
6.	CO2	Implementation of TSP using heuristic approach
7.	CO3	Implementation of the problem-solving strategies: either using Forward Chaining or Backward Chaining
8.	CO3	Create a knowledgebase consisting of first order logic statements and prove the given query using forward reasoning.
9.	CO4	Write a program to implement following: a. Program to calculate the probability of an event using its probability notation. b. Program to calculate marginal probability from a joint distribution. c. Program to represent a simple grid world environment similar to Wumpus World and perform basic actions.
10.	CO4	Implement Tic-Tac-Toe game using Python

Textbook:

1. Stuart J. Russell and Peter Norvig, Artificial Intelligence, 3rd Edition, Pearson,2015

Reference Book:

1. Elaine Rich, Kevin Knight, Artificial Intelligence, 3rd edition, Tata McGraw Hill,2013
2. George F Lugar, Artificial Intelligence Structure and strategies for complex, Pearson Education, 5th Edition, 2011

Online Resources:

1. Artificial Intelligence: https://nptel.ac.in/courses/106105077

Code: BCS403 **Course: Database Management System**
Credits: 4 **L:T:P - 3:0:2**
SEE: 100 Marks **CIE: 100 Marks**
SEE Hours: 3 **Max. Marks: 100**

Prerequisites if any	NIL
Learning objectives	1. Understand the fundamental concepts of databases, including database languages, architectures, and conceptual data modeling using entities and relationships. 2. Gain proficiency in relational database management systems, including the relational model, relational algebra, normalization, SQL, and transaction processing.

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Describe fundamental concepts of database management systems, including architecture, languages, and functionalities.	Understand
CO2	Design and implement database schemas using entities, relationships, and normalization techniques.	Apply
CO3	Demonstrate proficiency in SQL for data manipulation, retrieval, and management tasks.	Apply
CO4	Analyze and compare concurrency control mechanisms in relational databases and NoSQL databases, understanding their respective advantages and limitations.	Analyze

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-		-	-
CO2	3	3	2	-	-	-	-	-	-	-	-	-		2	2
CO3	2	2	2	-	3	-	-	-	-	-	-	2		2	3
CO4	2	-	-	-	3	-	-	-	-	-	-	3		2	2

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
	Module – 1: Introduction to Databases			
1.1	**Introduction to Databases:** Introduction, Characteristics of database approach, Advantages of using the DBMS approach, History of database applications.	2	-	0
1.2	**Overview of Database Languages and Architectures:** Data Models, Schemas, and Instances. Three schema architecture and data independence. Database languages, and interfaces, The Database System environment.	3	-	0
1.3	**Conceptual Data Modelling using Entities and Relationships:** Entity types, Entity sets and structural constraints, Weak entity types, ER diagrams, Specialization and Generalization	3	-	1
	Module – 2: Relational Databases			
2.1	**Relational Model:** Relational Model Concepts, Relational Model Constraints and relational database schemas, Update operations, transactions, and dealing with constraint violations.	3	-	0
2.2	**Relational Algebra:** Unary and Binary relational operations, additional relational operations (aggregate, grouping, etc.) Examples of Queries in relational algebra.	3	-	0
2.3	**Mapping Conceptual Design into a Logical Design:** Relational Database Design using	2	-	0

	ER-to-Relational mapping			
Module – 3: Normalization and SQL				
3.1	**Normalization:** Database Design Theory – Introduction to Normalization using Functional and Multivalued Dependencies: Informal design guidelines for relation schema, Functional Dependencies, Normal Forms based on Primary Keys, Second and Third Normal Forms, Boyce-Codd Normal Form, Multivalued Dependency and Fourth Normal Form, Join Dependencies and Fifth Normal Form.	5	-	1
3.2	**SQL:** SQL data definition and data types, Schema change statements in SQL, specifying constraints in SQL, retrieval queries in SQL, INSERT, DELETE, and UPDATE statements in SQL, Additional features of SQL.	3	-	2
Module – 4: SQL and Transactions				
4.1	**SQL:** Advanced Queries: More complex SQL retrieval queries, Specifying constraints as assertions and action triggers, Views in SQL.	3	-	2
4.2	**Transaction Processing:** Introduction to Transaction Processing, Transaction and System concepts, Desirable properties of Transactions, Characterizing schedules based on recoverability, Characterizing schedules based on Serializability, Transaction support in SQL.	5	-	1
Module – 5: Concurrency control and NoSQL Databases				
5.1	**Concurrency Control in Databases:** Two-phase locking techniques for Concurrency control, Concurrency control based on Timestamp ordering, Multiversion Concurrency control techniques, Validation Concurrency control techniques, Granularity of Data items and Multiple Granularity Locking.	4	-	1
5.2	**NoSQL Databases and Big Data Storage Systems:** Introduction to NOSQL Systems, The CAP Theorem, Document-Based NOSQL Systems and MongoDB, NOSQL Key-Value Stores, Column-Based or Wide Column NOSQL Systems.	4	-	2
	Total No. of Lecture Hours	**40**	-	-
	Total No. of Tutorial Hours		**00**	**-**
	Total No. of Practical Hours			**10**

PRACTICAL COMPONENT

Sl. No.	Experiments	COs
1.	Create a table called Employee & execute the following. Employee (EMPNO, ENAME, JOB, MANAGER_NO, SAL, COMMISSION) 1. Create a user and grant all permissions to the user. 2. Insert any three records in the employee table containing attributes. EMPNO, ENAME JOB, MANAGER_NO, SAL, COMMISSION and use rollback. Check the result. 3. Add primary key constraint and not null constraint to the employee table. 4. Insert NULL values to the employee table and verify the result.	CO3
2.	Create a table called Employee that contains attributes EMPNO, ENAME, JOB, MGR, SAL and execute the following. 1. Add a column commission with domain to the Employee table. 2. Insert any five records into the table. 3. Update the column details of job. 4. Rename the column of Employ table using alter command. 5. Delete the employee whose EMPNO is 105.	CO3
3.	Queries using aggregate functions (COUNT, AVG, MIN, MAX, SUM), Group by, Orderby. Employee (E_id, E_name, Age, Salary) 1. Create Employee table containing all Records E_id, E_name, Age, Salary. 2. Count number of employee names from employee table. 3. Find the Maximum age from the employee table. 4. Find the Minimum age from the employee table. 5. Find salaries of employees in Ascending Order.	CO3

	6. Find grouped salaries of employees.	
4.	Create a row level trigger for the customers table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table. This trigger will display the salary difference between the old & new Salary. CUSTOMERS (ID, NAME, AGE, ADDRESS, SALARY)	CO4
5.	Create cursor for Employee table and extract the values from the table. Declare the variables, Open the cursor, and extract the values from the cursor. Close the cursor. Employee (E_id, E_name, Age, Salary)	CO4
6.	Install an Open-Source NoSQL Data base MongoDB & perform basic CRUD (Create, Read, Update & Delete) operations. Execute MongoDB basic Queries using CRUD operations.	CO5
7.	Project-based Experiment: The project should use all the database concepts covered in theory and laboratory sessions. Students can integrate other relevant concepts/technologies as required.	CO1, CO2, CO3, CO4.

Textbook:

1. Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.

Reference Book:

1. Database management systems, Ramakrishnan, and Gehrke, 3rd Edition, 2014, McGraw Hill.

Online Resources:

1. MIT OpenCourseWare Course Link: https://ocw.mit.edu/courses/6-830-database-systems-fall-2010/
2. IIT Kharagpur Course Link: https://cse.iitkgp.ac.in/~pabitra/course/dbms/dbms_new.html
3. NPTEL Course Link: https://onlinecourses.nptel.ac.in/noc22_cs91/preview

ESTD : 1946

Analysis & Design of Algorithms Lab		Semester	4
Course Code	**BCSL404**	CIE Marks	50
Teaching Hours/Week (L:T:P)	0:0:2	SEE Marks	50
Credits	01	Exam Hours	2
Examination type (SEE)	Practical		

Course outcomes:

At the end of the course the student will be able to:

1. Develop programs to solve computational problems using suitable algorithm design strategy.
2. Compare algorithm design strategies by developing equivalent programs and observing runningtimes for analysis (Empirical).
3. Make use of suitable integrated development tools to develop programs
4. Choose appropriate algorithm design techniques to develop solution to the computational andcomplex problems.
5. Demonstrate and present the development of program, its execution and running time(s) and record the results/inferences.

Sl.No	CO's	Experiments
1	CO1 to CO5	Design and implement C/C++ Program to sort a given set of n integer elements using Selection Sort method and compute its time complexity. Run the program for varied values of n> 5000 and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
2	CO1 to CO5	Design and implement C/C++ Program to obtain the Topological ordering of vertices in a given digraph.
3	CO1 to CO5	Design and implement C/C++ Program to sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of n> 5000, and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
4	CO1 to CO5	Design and implement C/C++ Program to sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of n> 5000 and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
5	CO1 to CO5	Design and implement C/C++ Program to solve 0/1 Knapsack problem using Dynamic programming method.
6	CO1 to CO5	a. Design and implement C/C++ Program to solve All-Pairs Shortest Paths problem using Floyd'salgorithm. b. Design and implement C/C++ Program to find the transitive closure using Warshal's algorithm.

7	CO1 to CO5	Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Prim's algorithm.
8	CO1 to CO5	Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm.
9	CO1 to CO5	Design and implement C/C++ Program to find shortest paths from a given vertex in a weighted connected graph to other vertices using Dijkstra's algorithm.
10	CO1 to CO5	Design and implement C/C++ Program for N Queen's problem using Backtracking.

Suggested Learning Resources:

Textbook

1. Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian), 2017, Pearson.

2. Virtual Labs (CSE): http://cse01-iiith.vlabs.ac.in/

ESTD : 1946

Code: BCS405A **Course: Discrete Mathematical Structures**
Credits: 3 **L:T:P - 2:2:0**
SEE: 100 Marks **CIE: 50 Marks**
SEE Hours: 3 **Max. Marks: 100**

Prerequisites if any	Basics of number system, Mathematics
Learning objectives	1. To solve problems using concepts of Functions. 2. Solve problems using Relations and its properties. 3. To introduce Generating Functions and Recurrence Relations 4. To introduce concepts and properties of Graphs 5. To introduce the concepts of Trees and its properties.

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Apply the concepts and properties of Functions and Relations in solving problem.	Apply
CO2	Solve problems using Recurrence Relations and Study its applications in computers.	Apply
CO3	Solve problems using concepts of graphs and analyze its real-world applications.	Apply
CO4	Synthesis tree structure paradigm.	Analyze

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	3	3	2	2	-	-	-	-	2	1	2	2		1	2
CO2	3	3	2	2	-	-	-	-	2	1	2	2		2	3
CO3	3	3	2	2	-	-	-	-	2	1	1	2		2	3
CO4	3	3	2	3	-	-	-	-	2	1	3	2		2	3

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
	Module – 1: Functions			
1.1	**Functions:** Cartesian Products and Relations	2	0	0
1.2	Plain and One-to-One, Onto Functions	1	1	0
1.3	The Pigeonhole Principle	1	1	0
1.4	Function Composition and Inverse Functions	1	1	0
	Module – 2: Relations			
2.1	**Relations:** Properties of Relations,	1	1	0
2.2	Computer Recognition – Zero-One Matrices and Directed Graphs	2	1	0
2.3	Partial Orders – Hasse Diagrams.	2	1	0
	Module – 3: Recurrence Relations			
3.1	**Recurrence Relations:** First order linear recurrence relations,	2	0	0
3.2	The Second order linear homogeneous recurrence relation with constant coefficients	1	1	0

3.3	Non Homogeneous recurrence relation	2	1	0
Module – 4: Graph Theory and Applications				
4.1	**Graph Theory and Applications**: Definitions and Examples Sub graphs, Complements	1	1	0
4.2	Graph Isomorphism, Vertex Degree, Euler Trails and Circuits	1	0	0
4.3	Planar Graphs	1	1	0
4.4	Hamilton Paths and Cycles	1	1	0
4.5	Graph Coloring, and Chromatic Polynomials	1	1	0
Module – 5: Trees				
5.1	**Trees:** Definitions, Properties, and Examples	2	1	0
5.2	Rooted Trees	1	1	0
5.3	Trees and Sorting	1	1	0
5.4	Weighted Trees and Prefix Codes	1	0	0
	Total No. of Lecture Hours	**25**	-	-
	Total No. of Tutorial Hours		**15**	**-**
	Total No. of Practical Hours			**00**

Textbook:

1. **Discrete and Combinatorial Mathematics,** Ralph P. Grimaldi, 5th Edition, PHI/Pearson Education, 2004.

Reference Book:

1. **Handbook of discrete and combinatorial mathematics**, Kenneth H.Rosen, John G.Michels.
2. **Mathematics of Computer Science**, Prof. Albert R.Meyer, MIT Open Course Ware.
3. **Concrete Mathematics: A foundation for computer science**, Ronald L.Graham, Donald Ervin Knuth, Oren Patashnik
4. **Graph Theory with Applications to Engineering and Computer Science** by NarsinghDeo, Prentice-Hall, 2004

ESTD : 1946

Code: BDSL456C **Course: MERN (MongoDB, Express, React, Node)**
Credits: 1 **L: T: P – 0:0:2**
SEE: 50 Marks **CIE: 50 Marks**
SEE Hours:02 **Max. Marks: 100**

Prerequisites if any	Fundamentals of Web development and Client server technology
Learning objectives	• To gain proficiency in applying advanced Internet technology scripting languages, tools used in client server side for understanding, analyzing and applying. • To gain a comprehensive understanding of different internet technology tools like Express, React, Node and MongoDB which enables students to analyze, compare and apply them effectively in various problem solving contexts

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Apply the fundamentals of MongoDB, such as data modelling, CRUD operations, and basic queries to solvegiven problem.	Understanding
CO2	Use constructs of Express.js, including routing, software and constructing RESTful APIs to solve real worldproblems	Apply
CO3	Develop scalable and efficient RESTful APIs using NodeJS.	Apply
CO4	Develop applications using React, including components, state, props, and JSX syntax	Analyze

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	2	1	-	-	2	-	1	1	-	-	-	3		-	2
CO2	2	1	-	-	2	-	1	1	-	-	-	3		-	2
CO3	2	1	-	-	1	-	1	1	-	-	-	3		-	2
CO4	2	1	-	-	1	-	1	1	-	-	-	3		-	2

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

SL. No.	CO	Experiments	No. of Practical Hours
1	CO1	a.Using MongoDB, create a collection called transactions in database usermanaged (drop if it already exists) and bulk load the data from a json file, transactions.json b. Upsert the record from the new file called transactions_upsert.json in Mongodb.	2
2	CO1	Query MongoDB with Conditions: [Create appropriate collection with necessary documents to answer the query] a. Find any record where Name is Somu b. Find any record where total payment amount (Payment.Total) is 600. c. Find any record where price (Transaction.price) is between 300 to 500. d. Calculate the total transaction amount by adding up Payment.Total in all records.	2
3	CO1	Implement all CRUD operations on a File System using Node JS	1

4	CO2	a. Write a program to check request header for cookies. b. write node.js program to print the a car object properties, delete the second property and get length of the object.	2
5	CO2	Develop an authentication mechanism with email_id and password using HTML and Express JS (POST method)	2
6	CO2	Develop two routes: find_prime_100 and find_cube_100 which prints prime numbers less than 100 and cubes less than 100 using Express JS routing mechanism	2
7	CO1	. Read the data of a student containing usn, name, sem, year_of_admission from node js and store it in the MongoDB . For a partial name given in node js, search all the names from MongoDB student documents created in Question(a)	2
8	CO3	Develop the application that sends fruit name and price data from client side to Node.js server using Ajax	2
9	CO4	Develop a React code to build a simple search filter functionality to display a filtered list based on the search query entered by the user.	2
10	CO4	Develop a React code to collect data from rest API.	1
		Total No. of Practical Hours	**18**

Textbook:

1. Vasan SubramanianPro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Apress; 1st ed. edition (1 April 2017)
2. Eddy Wilson Iriarte Koroliova,MERN Quick Start Guide, Packt Publishing (31 May 2018)

Online Resources:

1. https://www.geeksforgeeks.org/mern-stack/
2. https://blog.logrocket.com/mern-stack-tutorial/

ESTD : 1946

Course Code:	**BBOK407**	**Course:**	**Biology for Engineers**
Credits:	**2**	**L:T:P:S**	**2:0:0:0**
SEE:	**50% Marks**	**CIE:**	**50% Marks**
SEE Hours:	**2 Hrs**	**Max. Marks:**	**100 (50+50)**

Prerequisites if any	None
Learning objectives	1. Review the basics of cell biology and role of biomolecules. 2. Elucidate the significance of Biomechanics and Biomaterials

Course Outcomes:

On the successful completion of the course, the student will be able to

	Course Outcomes	Bloom's level
CO1	Explain the fundamentals of Life, Evolution, Biomolecules, Cell Biology & Genetics	Understand
CO2	Outline the physical principles regulating the systems of the human bodies	Understand
CO3	Describe the impact of Biomaterials on the fields of Engineering & Medicine	Understand
CO4	Summarize tissue engineering and clinical applications of materials	Understand

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	1	1	-	1	-	3	2	-	-	-	-	1		-	-
CO2	1	1	-	1	-	3	2	-	-	-	-	1		-	-
CO3	1	1	-	1	-	3	2	-	-	-	-	1		-	-
CO4	1	1	-	1	-	3	2	-	-	-	-	1		-	-

Strong: 3 Medium: 2 Low: 1

Course Content

	Module - 1	No. of Lecture Hours	No. of Tutorial Hours	Self-Learning Hours
1.1	**Cell basic unit of life:** Introduction, Origin, and evaluation of life. Structure and functions of a cell. Stem cells and their application.	2	-	-
1.2	Biomolecules: Properties and functions of Carbohydrates, Nucleic acids, proteins, lipids. Importance of special biomolecules: Properties and functions of enzymes, vitamins, and hormones.	3	-	-
	Module - 2			
2.1	**Application of biomolecules:** Carbohydrates in cellulose-based water filters production, PHA and PLA in bioplastics production, Nucleic acids in vaccines and diagnosis,	2	-	-
2.2	Proteins in food production, lipids in biodiesel and detergents production, Enzymes in biosensors fabrication, food processing, detergent formulation, and textile processing.	3	-	-
	Module - 3			
3.1	**Adaptation of anatomical principles for bioengineering design:** Brain as a CPU system, Eye as a Camera system, Heart as a pump system.	3	-	-
3.2	Lungs as purification system, Kidney as a filtration system.	2	-	-
	Module - 4			
4.1	**Nature-bioinspired materials and mechanisms:** Echolocation, Photosynthesis, Bird flying, Lotus leaf effect, Plant burrs, Shark skin, Kingfisher beak.	3	-	-
4.2	Human Blood substitutes - haemoglobin-based oxygen carriers (HBOCs) and perfluorocarbons (PFCs).	2	-	-
	Module - 5			
5.1	**Trends in bioengineering:** Muscular and Skeletal Systems as scaffolds, and tissue engineering, Bioprinting techniques and materials.	3	-	-
5.2	Electrical tongue and electrical nose in food science, DNA origami and	2	-	-

	Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis, Bio concrete, Bioremediation, Biomining.			
	Total No. of Lecture Hours	25	-	-
	Total No. of Tutorial Hours		-	-
	Total No. of Self learning Hours			-

Detailed Lesson Plan:

Sr No. of Module	Number of related learning Objectives	Weeks/ Dates	Online Mode		ICT Tool/ Platform/ LMS	Face-to-face Mode	
			Resource (OER/ URL/ IM/ CP)	Activity (Describe activity in detail)		Resource (OER/ URL/ IM/ CP)	Activity
1.1	1	1	Refer succeeding section titled ' Online Resources'	-	PPT, Smart Board, Moodle	-	Student Led Discussions & Presentations
1.2	1	1		-		-	
2.1	1	1		-		-	
2.2	1	1		-		-	
3.1	1	2		-		-	
3.2	1	2		-		-	
4.1	2	1		-		-	
4.2	2	1		-		-	
5.1	2	2		-		-	
5.2	2	2		-		-	

Assessment Pattern:

Bloom's level	Continuous Internal Examination			End Semester Examination
	Test 1	Test 2	Assignment/Quiz/AAT	
Remember	√	√	√	√
Understand	√	√	√	√
Apply				
Analyze				
Evaluate				
Create				

Text Books:

1. Biology for Engineers by G. K. Suraishkumar; Oxford University Press, 2019, First Edition

Reference Books:

1. Introductory Biomechanics: From Cells to Organisms by C. Ross Ethier and Craig A. Simmons; Cambridge University Press, 2012, Online Edition
2. Introduction to Biomaterials: Basic Theory with Engineering Applications, J. L. Ong, Mark R. Appleford, Gopinath Mani, Cambridge University Press, 2014, First Edition
3. Biology for Engineers, Rajendra Singh C and Rathnakar Rao N, Rajendra Singh C and Rathnakar Rao, N Publishing, Bengaluru, 2023.
4. Human Physiology, Stuart Fox, Krista Rompolski, McGraw-Hill eBook. 16th Edition, 2022
5. Biology for Engineers, Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jaganthan M.K., Tata McGraw-Hill, New Delhi, 2012.

6. Biology for Engineers, Arthur T. Johnson, CRC Press, Taylor and Francis, 2011
7. Biomedical Instrumentation, Leslie Cromwell, Prentice Hall 2011.
8. Biology for Engineers, Sohini Singh and Tanu Allen, Vayu Education of India, New Delhi, 2014.
9. Biomimetics: Nature-Based Innovation, Yoseph Bar-Cohen, 1st edition, 2012, CRC Press.
10. Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies, D. Floreano and C. Mattiussi, MIT Press, 2008.
11. Bioremediation of heavy metals: bacterial participation, by C R Sunilkumar, N Geetha A C Udayashankar Lambert Academic Publishing, 2019.
12. 3D Bioprinting: Fundamentals, Principles and Applications by Ibrahim Ozbolat, Academic Press, 2016.
13. Electronic Noses and Tongues in Food Science, Maria Rodriguez Mende, Academic Press, 2016

Online Resources:

1. NOC: Biology for engineers and other non-biologists, IIT Madras; Dr. Madhulika Dixit, Prof. G.K. Suraishkumar, https://nptel.ac.in/courses/121106008
2. Introduction To Biological Engineering Design, MIT Open Courseware, https://ocw.mit.edu/courses/20-020-introduction-to-biological-engineering-design-spring-2009
3. Introduction To Bioengineering, MIT Open Courseware, https://ocw.mit.edu/courses/20-010j-introduction-to-bioengineering-be-010j-spring-2006

ESTD : 1946

Course Code: BUHK408
Credits: 1
SEE: -

Course Name: Universal Human Values and Professional Ethics
L:T:P - 1:0:0
CIE: 50 Marks

Prerequisites if any	Nil
Learning objectives	1. To help the students appreciate the essential complementarily between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity, which are the core aspirations of all human beings. 2. To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. 3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behavior and mutually enriching interaction with nature

Course Outcomes:
On the successful completion of the course, the student will be able to

COs		**Bloom's level**
CO1	To understand the core aspirations of all human beings	Understand
CO2	To gain the universal human values and movement towards value-based living in a natural way	Apply
CO3	To fulfilling the human behavior and mutually enriching interaction with nature	Apply

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2
CO1	-	-	-	-	-	3	1	2	3	3	1	2		-	-
CO2	-	-	-	-	-	3	2	2	3	3	1	2		-	-
CO3	-	-	-	-	-	3	2	2	3	3	1	2		-	-

S – Strong (3) M – Medium (2) L – Low (1)

Course Structure

Module – 1: Introduction - Need, Basic Guidelines, Content and Process for Value Education		**No. of Lecture Hours**	**No. of Tutorial Hours**
1.1	Understanding the need, basic guidelines, content and process for Value Education	1	Nil
1.2	Self-Exploration–what is it? - its content and process; 'Natural Acceptance' and Experiential Validation- as the mechanism for self-exploration	1	Nil
1.3	Right understanding, Relationship and Physical Facilities- the basic requirements for fulfillment of aspirations of every human being with their correct priority	1	Nil
1.4	Method to fulfill the above human aspirations: understanding and living in harmony at various levels . Practice session	2	Nil
Module – 2: Understanding Harmony in Myself, Family, Society and Human Relationship			
2.1	Understanding human being as a co-existence of the sentient 'I' and the material 'Body', Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha	1	Nil
2.2	Understanding Harmony in the family – the basic unit of human interaction. Understanding values in human-human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship	1	Nil

2.3	Understanding the meaning of Vishwas and Samman; Difference between intention and competence; respect and differentiation ; Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astitva as comprehensive Human Goals	1	Nil
2.4	Visualizing a universal harmonious order in society- Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha)- from family to world family! Practice session	2	Nil
Module – 3: **Understanding Harmony in the Nature, Existence and Implications of the all Holistic on Professional Ethics**			
3.1	Understanding the harmony in the Nature, Interconnectedness and mutual fulfillment among the four orders of nature recyclability and self-regulation in nature	1	Nil
3.2	Understanding Existence as Co-existence (Sah-astitva) of mutually interacting units in all-pervasive space.	1	Nil
3.3	Competence in professional ethics: a) Ability to utilize the professional competence for augmenting universal human order b) Ability to identify and develop appropriate technologies and management patterns for above production systems.	1	Nil
3.4	Strategy for transition from the present state to Universal Human Order: a) At the level of individual: as socially and ecologically responsible engineers, technologists and managers b) At the level of society: as mutually enriching institutions and organizations. Practice session	2	Nil
	Total No. of Lecture Hours	15	
	Total No. of Tutorial Hours		Nil

Guidelines and Content for Practice Sessions

Practice Session 1:

Introduce yourself in detail. What are the goals in your life? How do you set your goals in your life? How do you differentiate between right and wrong? What have been your achievements and shortcomings in your life? Observe and analyze them.

Expected outcome: The students start exploring themselves; get comfortable to each other and to the teacher and start finding the need and relevance for the course.

Practice Sessions 2:

1. a. Observe that any physical facility you use, follows the given sequence with time: Necessary & tasteful→ unnecessary & tasteful → unnecessary & tasteless → intolerable b. In contrast, observe that any feeling in you is either naturally acceptable or not acceptable at all. If naturally acceptable, you want it continuously and if not acceptable, you do not want it any moment!

2. List down all your activities. Observe whether the activity is of 'I' or of Body or with the participation of both 'I' and Body.

Expected outcome:

1. The students are able to see that all physical facilities they use are required for a limited time in a limited quantity. Also they are able to see that in case of feelings, they want continuity of the naturally acceptable feelings and they do not want feelings which are not naturally acceptable even for a single moment.

2. The students are able to see that activities like understanding, desire, thought and selection are the activities of 'I' only, the activities like breathing, palpitation of different parts of the body are fully the activities of the body with the acceptance of 'I' while the activities they do with their sense organs like hearing through ears, seeing througbody.es, sensing through touch, tasting through tongue and smelling through nose or the activities they do with their work organs like hands, legs etc. are such activities that require the participation of both 'I' and body

Practice Session 3:

Form small groups in the class and in that group initiate dialogue and ask the eight questions related to trust. The eight questions are:

1a. Do I want to make myself happy?

2a. Do I want to make others happy?
3a. Does the other want to make him happy?
4a. Does the other want to make me happy?
What is the answer? Intention (Natural Acceptance)
1b. Am I able to make myself always happy?
2b. Am I able to make others happy?
3b. Is the other able to make him always happy?
4b. Is the other able to make me always happy?
What is the answer? Competence
Let each student answer the questions for himself and everyone else. Discuss the difference between intention and competence. Observe whether you evaluate your intention & competence as well as the others' intention & competence.

Expected outcome:

The students are able to see that the first four questions are related to our Natural Acceptance i.e. Intention and the next four to our Competence. They are able to note that the intention is always correct, only competence is lacking! We generally evaluate ourselves on the basis of our intention and others on the basis of their competence! We seldom look at our competence and others' intention as a result we conclude that I am a good person and other is a bad person.

Textbooks:

1. R.R Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and professional Ethics, Excel books, New Delhi, 2010, ISBN 978-8-174-46781-2.

References:

1. IIT Delhi, Modern Technology – the Untold Story

ESTD : 1946

Course Code: BNSK459
Credits: Zero
SEE: NA
SEE Hours: NA

Course: National Service Scheme (NSS)
L:T:P 0:0:2
CIE: 100 Marks
Max. Marks: 100

Prerequisites if any	1. Students should have a service oriented mind set and social concern. 2. Students should have dedication to work at any remote place, anytime with available resources and proper time management for the other works. 3. Students should be ready to sacrifice some of the time and wishes to achieve service oriented targets on time
Learning objectives	1. Understand the community in which they work 2. Identify the needs and problems of the community and involve them in problem-solving 3. Develop among themselves a sense of social & civic responsibility & utilize their knowledge in finding practical solutions to individual and community problems 4. Develop competence required for group-living and sharing of responsibilities & gain skills in mobilizing community participation to acquire leadership qualities and democratic attitudes 5. Develop capacity to meet emergencies and natural disasters & practice national integration and social harmony

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes
CO1	Understand the importance of his / her responsibilities towards society.
CO2	Analyse the environmental and societal problems/issues and will be able to design solutions for the same
CO3	Evaluate the existing system and to propose practical solutions for the same for Sustainable development and Implement government or self-driven projects effectively in the field.

Mapping with POs and PSOs:

COs	PO1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2		PSO 1	PSO 2	PSO 3	PSO 4
CO1	-	-	3	-	-	-	2	2	-	-	-	1		To be identified for each branch by Course Instructor			
CO2	-	3	-	-	-	-	2	-	-	-	-	-					
CO3	-	-	-	-	-	2	3	-	-	-	-	-					

Mapping Strength: 2 Strong: 3 Medium –2 Low – 1

Course Structure

		No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
	List of Events			
1	Organic farming, Indian Agriculture (Past, Present and Future) Connectivity for marketing.			
2	Waste management– Public, Private and Govt organization, 5 R's.			
3	Setting of the information imparting club for women leading to contribution in social and economic issues.			
4	Water conservation techniques – Role of different stakeholders– Implementation.			
5	Preparing an actionable business proposal for enhancing the village income and approach for implementation.			
6	Helping local schools to achieve good results and enhance their enrolment in Higher/ technical/ vocational education.			
7	Developing Sustainable Water management system for rural areas and implementation approaches.			

8	Contribution to any national level initiative of Government of India. Digital India, Skill India, Swachh Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc.			
9	Spreading public awareness under rural outreach programs.(minimum5 programs)			
10	Social connect and responsibilities.			
11	Plantation and adoption of plants.			
12	Govt. school Rejuvenation and helping them to achieve good infrastructure			
13	Sustainable agriculture practices - Organic farming, Agroforestry and crop rotation.			
14	Rural finance – role of local bodies, need for agricultural finance and sources of agricultural finances.			
15	Strategies for the development of rural markets and emerging issues in rural marketing			
16	Rural energy system – conventional and non-conventional, Rural electrification-policies, achievements and targets.			
17	Livestock economies - fishery and poultry development, forestry and horticulture.			
18	Role of NGO's in rural development, the role of voluntary organization in India's development processes.			
19	Issues in rural industrialization and development of agro-based industries, rural non-farm sector			
	Total No. of Lecture Hours	-	-	-
	Total No. of Tutorial Hours		-	-
	Total No. of Practical Hours			**26**

ONENSS – CAMP @ College /University /State or Central Govt Level /NGO's /General Social Camps

- Students have to take up anyone activity on the above said topics and have to prepare content for awareness and technical contents for implementation of the projects and have to present strategies for implementation of the same. Compulsorily students have to attend one camp.

- CIE will be evaluated based on their presentation, approach and implementation strategies.

Suggested Learning Resource:

1. NSS Course Manual, Published by NSS Cell, VTU Belagavi.

Course Code: BPEK459 **Course: Physical Education (Sports & Athletics) – II**
Credits: Zero **L:T:P 0:0:2**
SEE: NA **CIE: 100 Marks**
SEE Hours: NA **Max. Marks: 100**

Course Outcomes: At the end of the course, the student will be able to

1. Understand the ethics and moral values in sports and athletics
2. Perform in the selected sports or athletics of student's choice.
3. Understand the roles and responsibilities of organisation and administration of sports and games.

Module IV : Ethics and Moral Values **5 Hours**
A. Ethics in Sports B. Moral Values in Sports and Games
Module V : Specific Games (Any one to be selected by the student) **20 Hours** A. Volleyball – Attack, Block, Service, Upper Hand Pass and Lower hand Pass. B. Throwball – Service, Receive, Spin attack, Net Drop & Jump throw. C. Kabaddi – Hand touch, Toe Touch, Thigh Hold, Ankle hold and Bonus. D. Kho-Kho – Giving Kho, Single Chain, Pole dive, Pole turning, 3-6 Up. E. Table Tennis – Service (Fore Hand & Back Hand), Receive (Fore Hand & Back Hand), Smash. F. Athletics (Track / Field Events) – Any event as per availability of Ground.
Module VI: Role of Organisation and administration **5 Hours**

Scheme and Assessment for auditing the course and Grades:

Sl. No.	Activity	Marks
1.	Participation of student in all the modules	20
2.	Quizzes – 2, each of 15 marks	30
3.	Assignment	50
	Total	**100**

Course Code: BYOK459 **Course: Yoga**
Credits: Zero **L:T:P 0:0:2**
SEE: NA **CIE: 100 Marks**
SEE Hours: NA **Max. Marks: 100**

Prerequisites if any	None
Learning objectives	1. To enable the student to have good health and mental hygiene. 2. To possess emotional stability 3. To integrate moral values 4. To attain higher level of consciousness.

Course Outcomes:

On successful completion of the course, the student will be able to:

Course Outcomes		Bloom's level
CO1	Understand the meaning of Yoga, its origin, history, development and importance.	Understand
CO2	Perform various Surya namaskar and able to Teach its benefits	Apply
CO3	Perform various asanas and able to Teach its benefits	Apply
CO4	Understand Benefits of Yoga on fitness and health	Apply

Course Content

	Module – 1	No. of Lecture and Practical Hours	No. of Tutorial Sessions
1.1	Role of yoga in controlling diseases	1	-
1.2	Patanjali's Ashtanga Yoga, its need and importance.	1	-
1.3	Yama :Ahimsa, satya, asteya, brahmacarya, aparigraha	1	-
1.4	Niyama :shoucha, santosh, tapa, svaadhyaya, Eshvarapranidhan	1	-
	Module – 2		
2.1	Warmup Exercise	2	-
2.2	Yoga jogging	2	-
2.3	Suryanamaskar12 count- 4 rounds of practice	2	-
2.4	Asana its meaning by name, technique, precautionary measures and benefits of each asana.	2	-
2.5	Sitting: 1.Sukhasana 2. Paschimottanasana 3.Bharadwajasana	2	-
2.6	Standing: 1. Ardhakati Chakrasana 2. Parshva Chakrasana	2	-
2.7	Prone line: 1.Makarasana 2.Dhanurasana	2	
2.8	Supine line 1. Halasana 2. Karna Peedasana	2	
	Module – 3		
3.1	Pranayama – Suryanuloma, Chandranuloma,	2	-
3.2	Suryabhedana,Chandra Bhedana,Nadishodhana	2	-
	Total No. of Lecture and practical Hours	**24**	-
	No. of Tutorial Sessions		**Nil**

Detailed Lesson Plan

Sl. No. of Module	Number of related learning Objectives	Weeks/ Dates	Online Mode		ICT Tool/ Platform/ LMS	Face-to-face Mode		Duration in Minutes
			Resource (OER/ URL/ IM/ CP)	Activity (Describe activity in detail)		Resource (OER/ URL/ IM/ CP)	Activity	
1.1	1 & 4	W 1			PPT, SMART BOARD, MOODLE		Explanation	60
1.2	1 & 4	W 1					Explanation	60
1.3	1 & 4	W 2					Explanation	60
1.4	1 & 4	W 2					Explanation	60
1.5	1 & 4	W 3					Explanation	60
1.6	1 & 4	W 3					Explanation	60
2.1	2 & 4	W 4					Practicing Surya namaskar	60
2.2	2 & 4	W 4					Practicing Surya namaskar	60
2.3	2 & 4	W 5					Practicing Surya namaskara	60
2.4	2 & 4	W 5					Practicing Surya namaskar	60
2.5	2 & 4	W 6					Practicing Surya namaskar	60
2.6	2 & 4	W 6					Practicing Asana	60
3.1	3 & 4	W 7					Practicing Asana	60
3.2	3 & 4	W 7					Practicing Asana	60
3.3	3 & 4	W 8					Practicing Asana	60
3.4	3 & 4	W 8					Practicing Asana	60
3.5	3 & 4	W 9					Practicing Asana	60
3.6	3 & 4	W 9					Practicing Asana	60
3.7	3 & 4	W10					Practicing Asana	60
3.8	3 & 4	W10					Practicing Asana	60
3.9	3 & 4	W 11					Practicing Asana	60
3.10	3 & 4	W11					Practicing Asana	60
3.11	3 & 4	W12					Practicing Asana	60
3.12	3 & 4	W 12					Practicing Asana	60

Assessment Pattern:

Bloom's level	Continuous Internal Examination			Semester End Examination
	Test 1	Test 2	Assignment	
Remember	-	-	-	-
Understand	✓	✓	✓	✓
Apply	✓	✓	✓	✓

Suggested Learning Resources:

Text Books:

1. Yogapravesha in Kannada by Ajitkumar
2. Light on Yoga by BKS Iyengar
3. Teaching Methods for Yogic practices by Dr. M L Gharote & Dr. S K Ganguly
4. Yoga Instructor Course hand book published by SVYASA University, Bengaluru
5. Yoga for Children –step by step – by Yamini Muthanna

Web links and Video Lectures (e-Resources): Refer links

1. https://youtu.be/KB-TYlgd1wE
2. https://youtu.be/aa-TG0Wg1Ls